A Pilot Study of Probing before Trusting Large Language Models in Self-Learning

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***Abstract*—A critical and general problem of large language models (LLMs) is that they may hallucinate, generating specious answers, especially when they interpret certain concepts expressed in the given natural language queries using incorrect (sub)domain knowledge they possess. Still, LLMs are widely used by people. To embrace LLMs in education, students can adopt them to seek quick feedback on their questions to enhance their inquisitive approaches to self-learning, while there is a guard against hallucination. This paper introduces *DomainProbe*, the first approach to domain-level hallucination detection that leverages metamorphic testing for addressing the test oracle problem to improve the trustworthiness of the feedback generated by LLMs. Given a question posed by a student, *DomainProbe* prompts the LLM to extract key topical terms from the question and provide explanations for each. The student then evaluates whether there is any inconsistent term-explanation pair. If such an inconsistency is identified, the corresponding answer for the question from the LLM is flagged as untrustworthy. We show *DomainProbe* to achieve promising results by evaluating it on MMLU, a widely used question-answer benchmark dataset. We further discuss our vision on the approach to promote students' learning objectives and outline future work for the metamorphic relations formulated in *DomainProbe*.**

***Index Terms*—AI hallucination detection, metamorphic testing, test oracle, educational technology, generative AI**

I. Introduction

The integration of generative artificial intelligence (GAI) in education has brought significant changes in content and recommendations. These platforms adopt AI technologies with the aim of providing high-quality content and timely responses to users. There is no exception for the contents on educational websites (e.g., Quora) for self-learning [1], [2]. Many students frequently use these platforms or websites to find answers to their queries.

Among AI technologies, large language models (LLMs) have recently gained considerable attention due to their ability to process natural language text and generate human-like text as output [3], [4]. In essence, LLMs generate output texts based on statistical probabilities and patterns to process and respond to queries. Recent research highlights the promise of using LLMs in producing accurate answers in various educational subjects, such as demonstrating their achieved student-level performance on standardized tests [5]. We agree that integrating LLMs in students' learning process could be transformational, potentially reshaping traditional learning methods and styles [6] and accelerating their pace of acquiring knowledge.

However, despite their potential, the current generation of LLMs, such as GPT-4o, incurs severe limitations [7]–[9]. There is still insufficient evidence to show that these LLMs possess the capability for deep semantic comprehension and human-level critical thinking. Thus, one of the main concerns is their paradoxical answers in response to out-of-ability queries provided by students, particularly in specialized STEM subject areas. For instance, previous studies [10] have shown that LLMs tend to produce plausible-sounding, factually incorrect, or nonsensical outputs when processing queries incorporated with concepts outside their embedded knowledge. This significant drawback is known as *AI hallucination* [11].

To address AI hallucination, most platforms or websites that provide content generated by GAI only state a disclaimer about possible incorrectness near their generated content. If students simply reuse the LLM-generated answers without their process of critical evaluation and reflection on them, they may only develop a superficial understanding of the topic, blocking them from engaging in higher-order thinking skills such as analysis, synthesis, and evaluation [12]. Thus, using the content generated by LLMs without a deeper understanding of the LLMs' reasoning processes can be detrimental to students' learning abilities [13].

From an education perspective, as described in Bloom's taxonomy of learning objectives [14], analyzing a topic in a subject is located higher than applying the topic in the taxonomy of "higher order of thinking". If students stop their learning process at the level of reusing the answers generated by LLMs as their answers to the original questions, they cannot acquire the skills to reach the next level of the learning objective (i.e., analysis) in the taxonomy. Although educators or AI technologies should not replace students in analyzing a topic in their self-learning processes, our insight is that software engineering technology can be employed to bridge the gap from the level of applying a topic to the level of analyzing that topic to assist students in reaching the analysis level more readily — by asking LLMs to explain topical terminologies and let students judge whether the explanations are consistent with the scopes of the topic that the original queries refer to.

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In software engineering (SE) research, determining the correctness of a program's output requires a test oracle. When a test oracle is difficult to obtain, as is often the case when students use LLMs to find answers to questions they don't know, this is known as the *test oracle problem*. One SE technology to address the test oracle problem is *metamorphic testing*, which has shown promise in addressing the AI reasoning problems [15]. Instead of checking each output against an (unknown) test oracle, metamorphic testing alleviates the test oracle problem by comparing the results of multiple outputs of the same program for multiple and related inputs. The key design element in metamorphic testing is the formulation of *metamorphic relations* (which is a kind of necessary property of the expected behavior of the program under test), which describes how the output of a program should be varied in response to certain transformations of the corresponding program input.[1] Nonetheless, in the context of the testing of LLMs, to the best of our knowledge, generic metamorphic relations are still unachievable.

Our proposed methodology to support student self-learning can be summarized as follows. Let Q denote a question query, f be an LLM, and $f(Q)$ be the content provided by f for answering Q. The methodology involves the following steps:

1) The question query Q is posed.
2) The answer $f(Q)$ is obtained.
3) A set of topical keywords $\langle k_1, \ldots, k_i, \ldots, k_n \rangle$ from Q.
4) The LLM f explains each topical keyword k_i (for $1 \leq i \leq n$) in the context of Q (denoted as $f(k_i \in Q)$).
5) The student reads the generated explanations on how the pair ($f(k_i \in Q)$, $f(Q)$) relates, and judges whether there is any inconsistency.
6) If the student identifies at least one inconsistency, the answer $f(Q)$ is marked as *untrustworthy*.

This methodology, named *DomainProbe*, actively engages the student in analyzing the consistency of the topical keywords in LLM generations, helping them learn to reason and relate the generated answers to their background knowledge. We have conducted an experiment to evaluate the efficacy and effectiveness of our proposed methodology. The experimental results show that the proposed methodology is able to detect inconsistencies that infer incorrect answers and achieves no false positive alarms.

This approach goes beyond an intermediate assessment of being knowledgeable to question domains as preliminary checks, focusing on the underlying cognitive processes of the LLMs. The falsity check result may imply that the AI-generated content would be produced due to AI hallucination. In this way, students can analyze the LLM capabilities and the reliability of AI-generated content and, therefore, strengthen their higher-order thinking skills.

The remainder of this paper is structured as follows: Section II provides background on LLMs, related software techniques, and learning objectives. Section III presents our methodology, followed by its evaluation in Section IV. We discussed the implications of our methodology in Section V. Finally, we conclude the work and suggest future research directions in Section VI.

II. Background

A. Large Language Models

Large language models [16] are a kind of GAI software designed to understand and generate human-like text. LLMs are trained on vast amounts of text data and utilize deep learning architectures, particularly transformer models, to process and generate natural language outputs. During training, optimization techniques such as gradient descent are employed to minimize the perplexity of the generated outputs, enabling the models to improve their performance iteratively.

Despite their impressive capabilities, LLMs are prone to AI hallucinations [10], which refers to when they generate outputs that are incorrect, misleading, or entirely fabricated, often presenting them with unwarranted confidence. AI hallucinations stem from limitations in insufficient training data, training data quality, and training adequacy. Users of LLMs generally have no indications to know whether a hallucination occurs.

B. Named Entitity Recognition

Named entity recognition (NER) [17] is a task of natural language processing (NLP) that involves identifying and classifying key entities in text into predefined categories. These categories typically include person names, organizations, locations, etc. Deep learning techniques have achieved significant performance in NER, often approaching high accuracy in standard benchmarks [18]. NER is often applied to extract relevant information from unstructured text and facilitate data analysis in areas like finance, healthcare, and academia.

C. Metamorphic Testing

Metamorphic testing [19] is a property-based software testing technique designed to address challenges in software verification, particularly the test oracle problem, which arises when it is difficult to determine whether the actual outputs of a program match the expected outcomes. Metamorphic relations are necessary properties of software under test based on the idea that if a specific change to the input, the output should also exhibit a corresponding change.

D. Taxonomy of Learning Objectives

Anderson et al. [14] introduce six levels of learning objectives shown in Fig. 1. Specifically, learning objectives from lower-order thinking to higher-order ones are knowing, understanding, applying, analyzing, evaluating, and creating. By taking higher levels of learning activities around these objectives, students can acquire higher-order thinking skills and gain a deeper understanding of desired learning outcomes.

[1]For instance, let sin(·) be a program that implements a mathematical sine function. It would be difficult first to find out the exact sine value for the input $\pi/1.23$ and use it as the expected value to check against the output of $sin(\cdot)$. A possible metamorphic relation is $sin(x) > sin(x/2)$ for $0 < x < \pi/2$. The metamorphic relation may not be able to detect all possible faults in $sin(\cdot)$, yet if any violation of the metamorphic relation is observed, $sin(\cdot)$ must have triggered its faults to produce failures in the output when processing inputs related to the given x.

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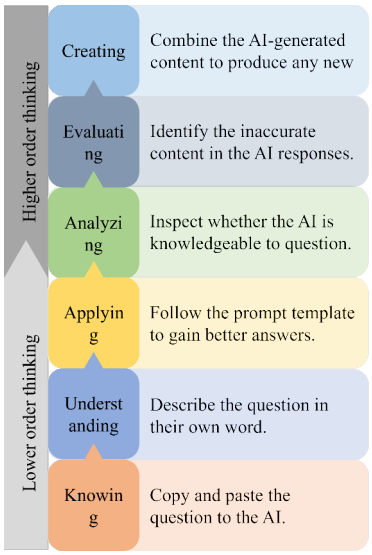


Fig. 1. Six levels of learning objectives proposed by Anderson et al. [14]. Instantiated examples of students using generative AI in this work.

III. Methodology

Fig. 2 depicts our proposed methodology of *DomainProbe*. The objective is to assess whether the LLM under query is capable of answering the question in a specific domain before trusting the LLM-generated answers.

A. Proposed Metamorphic Relations

The idea is to inspect the topical keywords in the question, which could be either the domain-specific terminologies or abbreviations. The LLM under query can produce the correct answer *if and only if* the following metamorphic relations (MRs) are not violated:

- MR1: The explanation of the term matches the correct knowledge that the question refers to.
- MR2: The full form of the abbreviation matches the true term that the question refers to.

> Example question for MR1: According to **Hooke's law**, if you hang by a tree branch and note how much it bends, then hanging with twice the weight produces ____.

For questions mentioning the topical keywords (in bold) explicitly, involving the knowledge behind these keywords is generally necessary in order to answer the questions correctly. In contrast, if the LLM under query does not explain the knowledge points well, it implies that such an LLM is not certified to provide the correct answer. An example can be found above.

> Example question for MR2: You are given a message (m) and its **OTP** encryption (c). Can you compute the OTP key from m and c?

For questions containing abbreviations, replacing the abbreviation with its full form does not change the semantics of the question itself. Moreover, it enables precise mentioning of the terminology in the context of the question. In contrast, if the LLM under query could not tell the correct full form of the abbreviation, it implies that the generated answer by such LLM may refer to other wrong terms in this generation process. As a result, the generated answer by such an LLM is not reliable. An example can be found in the above framebox.

These MRs are formulated as a checking mechanism to assist students in analyzing AI-generated content.

B. Probing Mechanism

Given a question Q for asking, the first step is identifying the named entities as topical keywords in the question context, producing a set of discovered entities E. For example, in Fig. 2, the question, *If* $f : \mathcal{R} \rightarrow \mathcal{R}$ *is a bounded function that is Lebesgue integrable, then which of the following must be true?* ..., contains one entity, *Lebesgue integrable*.

For each entity e_i in E, a $\texttt{check}(\cdot)$ function is performed on e_i based on the same LLM M under the context of Q. The *check* action returns a boolean predicate, indicating whether M satisfies the abovementioned MRs. Formally, the LLM M is deemed as *unknowledgeable* for answering Q if the following condition satisfies,

$$\exists\ e_i \in \mathsf{E}, s.t.\ \texttt{check}\,(\mathsf{M}, e_i, Q) = \mathsf{False} \tag{1}$$

If such an unknowledgeable case is detected, the AI-generated content by M for Q is marked as *untrustworthy*.

IV. Evaluation

To evaluate the effectiveness of our proposed methodology, we conduct a quantitative experiment on the proposed mechanism. Since this is a pilot study, we aim to answer the following research questions as an initial step.

1) RQ1: Can the proposed methodology detect any unknowledgeable case in popular LLM evaluation benchmarks?
2) RQ2: How effective is the proposed mechanism related to the correctness of AI-generated answers?

A. Subject AI models

GPT-4o model [20] is chosen in this evaluation due to its significant popularity during the experiment conducted. GPT-4o empowers the ChatGPT service[2] as the cutting-edge LLM model in 2024. In the education circumstances, many institutions like universities use Microsoft Azure OpenAI service[3] that generates text in dialogue format so as to facilitate their community to embrace the AI technology. Among the

[2]https://chat.openai.com
[3]https://learn.microsoft.com/en-us/azure/ai-services/openai/

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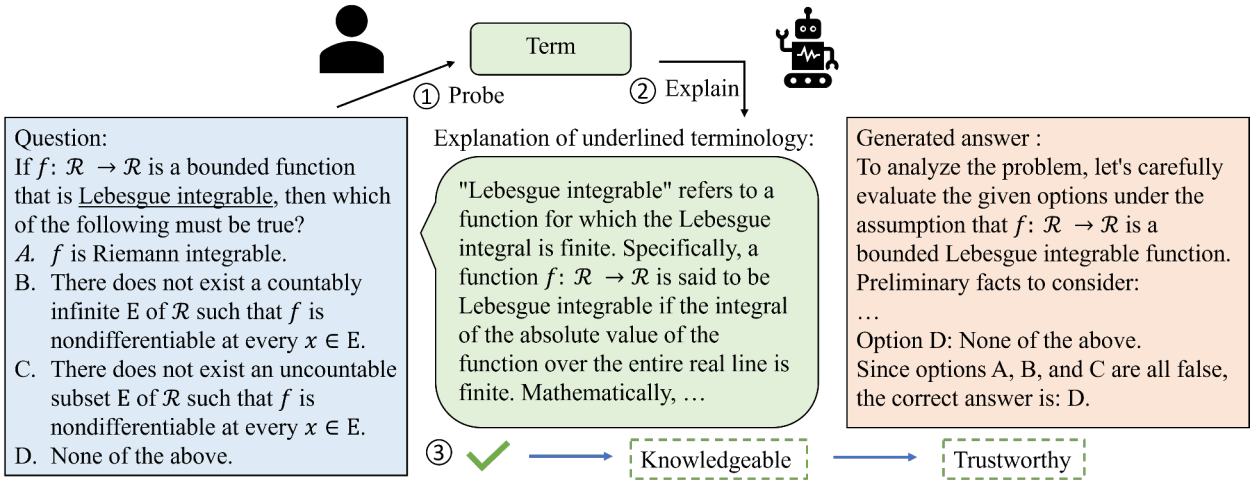


Fig. 2. The proposed probing before trusting methodology of *DomainProbe* in this work.

services, GPT-4o is the latest and most capable model that OpenAI and Azure provide. So, we choose the GPT-4o model as the subject AI model in our experiment.

B. Dataset

This study adopts a famous LLM benchmark dataset, Massive Multitask Language Understanding (MMLU) [21], to evaluate the proposed mechanism. The benchmark consists of 15908 multiple-choice questions, covering 57 subjects with different difficulty levels, such as "elementary", "high school", "college", or "professional", and drawing from real-world examinations [22].

Since it is still a pilot study, we constrain the evaluation scope to STEM subjects as a starting point and limit the number of questions to 50, as the evaluation requires human intervention to label the correctness of AI responses. The constructed dataset in this study can be found in Table I.

C. Experiment Design

Dataset construction. We downloaded the test set of the MMLU dataset from here[4]. We filtered out the questions that do not belong to STEM subjects. More specifically, the remaining questions are either math, physics, chemistry, biology, computer science, or engineering. Then, we randomly drew a question from the dataset and looked into it to ensure that the question under test contained at least one topical keyword in its subject. We underlined the keywords as the identified terms. The pairs of the question and its terms were then added to our constructed dataset as presented in Table I.

Probing LLMs. We follow the method as described in Section III-B to check whether the subject LLM is knowledgeable about the questions under test. The `check(·)` function is implemented as a prompt to query the LLM for the explanation of the underlined term under the question context. The prompt we developed for this study is shown below.

Prompt to Probe LLM

What is {{ term }} under the following question context? No need to answer the question. Just explain the term.

{{ question }}

We then evaluated the returned response manually by referring to Wikipedia to make the judgment according to (1). This process simulates the situation where students inspect the AI tools in their self-learning.

Acquiring answers. The MMLU dataset [21] provides the evaluation code [23] for OpenAI models. The evaluation constructs prompts using few-shot learning settings to guide the LLM in generating the answers to the multiple-choice questions. We adopted the provided evaluation code in our experiment and executed it to acquire the answers generated by the subject LLM.

D. Experiment Results

Table I presents the experiment results, where the first three columns present the dataset we constructed, the fourth column shows the results of the decision of knowledgeable or not, and the last column shows the correctness of the AI-generated answers with respective to the true correct answers provided by the MMLU dataset.

Although the MMLU dataset was published in 2021 and may have been used to train with its training set by commercial LLMs, the GPT-4o model released in 2024 cannot get full marks in the MMLU dataset. Among our dataset, the GPT-4o model fails on 3 out of 50 questions. The accuracy is 94%.

[4]https://people.eecs.berkeley.edu/∼hendrycks/data.tar

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TABLE I
DATASET & RESULT OF THE EXPERIMENT

id	subject	term	K? *	C? *
1	math	Fermat's theorem	✗	✗
2	math	eigenvector	✓	✓
3	math	Lebesgue integrable	✓	✓
4	math	hypothesis testing	✓	✓
5	math	correlation coefficient	✓	✓
6	math	P-value	✓	✓
7	physics	ALH84001	✓	✓
8	physics	Betelgeuse	✓	✓
9	physics	Ptolemy	✓	✓
10	physics	Cepheid	✓	✓
11	physics	Hooke's law	✓	✓
12	physics	Coulomb	✓	✓
13	physics	Huygens' principle	✓	✓
14	physics	Newton's law	✓	✓
15	chemistry	Hund's rule	✓	✓
16	chemistry	ideal gas law	✓	✓
17	biology	Barr body	✓	✓
18	biology	Brood	✓	✓
19	biology	Hamilton's rule	✓	✓
20	biology	Ferns	✓	✓
21	biology	Mouse sperm	✓	✗
22	biology	Charles Darwin	✓	✓
23	biology	Fungi	✓	✓
24	biology	Hardy-Weinberg equilibrium	✓	✓
25	biology	natural selection	✓	✓
26	biology	Zika fever	✓	✓
27	biology	Krebs cycle	✓	✓
28	biology	meiosis	✓	✓
29	computer	k-Means Clustering	✓	✓
30	computer	TCP protocol	✓	✓
31	computer	Internet Protocol	✓	✓
32	computer	Hash tables	✓	✓
33	computer	MTU	✓	✓
34	computer	greedy algorithm	✓	✓
35	computer	hash function	✓	✗
36	computer	Diffie Hellman method	✓	✓
37	computer	buffer overflow	✓	✓
38	computer	penetration testing	✓	✓
39	computer	OTP encryption	✓	✓
40	computer	remote Trojan	✓	✓
41	computer	SQL injection	✓	✓
42	computer	Wireless traffic Sniffing	✓	✓
43	computer	buffer overflow	✓	✓
44	computer	Nmap scan	✓	✓
45	computer	WPA	✓	✓
46	engineering	Barkhausen criterion	✓	✓
47	engineering	Potentiometer method	✓	✓
48	engineering	D-flip-flop	✓	✓
49	engineering	Piezoelectric effect	✓	✓
50	engineering	Bohr model	✓	✓

* "K" denotes knowledgeable, and "C" denotes correct.

Applying our methodology, we detected 1 exception for 50 questions (i.e., question id of #1). Specifically, the subject LLM explains the following for the term in question #1.

Fermat's Last Theorem is a much more famous result that remained unproven for over 350 years. It states: There are no three positive integers a, b, and c that satisfy the equation: $a^n + b^n = c^n$ *for any integer* $n > 2$.

where question #1 asks *Using Fermat's theorem, find the remainder of* 3^{47} *when it is divided by 23.*

Although the question was not stated clearly, the subject LLM referred to a wrong yet similar theorem (w.r.t to the theorem name). The true theorem that question #1 would like to refer to is Fermat's Little Theorem. Consequently, the generated answer of #1 was wrong.

Answering RQ1, the proposed mechanism is potentially able to detect some wrong references in the LLM's inherent generation process.

TABLE II
RESULT OF CONFUSION MATRIX

		C=Correct?	
	Population All=50	Positive (P)	Negative (N)
K=Knowledgeable?	True (T)	TP = 47	TN = 2
	False (F)	FP = 0	FN = 1

The results in Table I can also be regarded as a statistical classification problem [24]. The statistical result is summarized in Table II. TP cases are expected generally. TN cases mean that although LLMs know some concepts, they cannot generate correct answers. FP cases represent the negative effects of our methodology, where the LLM has no knowledge about core terminology, while it generates correct answers. FN cases indicate the exception cases to be detected in our study.

The experimental result shows that the proposed mechanism did not incur false positive (FP) cases in the study and could detect negative cases (i.e., FN). Answering RQ2, the proposed mechanism is effective for its promising performance.

V. DISCUSSION

As we instantiate the action examples in Fig. 1 for each level of Bloom's taxonomy of learning objectives [14], this work aims to promote students' critical thinking in order to match a higher level.

By applying our methodology, students could be able to detect some wrong references in the underlying process of the LLMs. After identifying such cases, they could describe the term more specifically (e.g., *Fermat's theorem* $\Rightarrow$ *Fermat's little theorem* in question #1) with some editing, they may be able to avoid the wrong reference by LLMs. In this way, students were promoted to the understanding level of the learning objective as they could describe the question in their own words. We believe that students adopting our methodology can analyze whether the LLM under use is feasible to answer their questions, and if not, the negative feedback can also guide them to revise their prompts to provide more relevant information to the LLM for more accurate answers.

Although we present to evaluate whether our methodology can detect some failures in this work, the application of our methodology could be generalized more. Students may also perform the checkings as we proposed to the AI-generated content to evaluate how far the content could be trusted. The checking could happen after the AI generation process. We believe that there could be more checking we could explore to strengthen the students' critical thinking in this direction.

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VI. Conclusion And Future Work

Our pilot study proposes a probing before trusting methodology for self-learning. The methodology is realized as a checking mechanism for inspecting LLM's capabilities. The evaluation showcases promising results of the mechanism. Students could adopt this methodology in their self-learning to promote their learning objectives. In the future, we plan to extend this mechanism into an automated checking middleware for existing generative AI tools.

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